PHY 1203 General Physics II
Modern Physics

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Modern Physics (18 hrs)

- Introductory Quantum Physics (4 hrs)
- Atomic Physics (4 hrs)
- Nuclear Physics (8 hrs)
- Elementary Particles (2 hrs)

Introductory Quantum Physics

Experiments led the development of Quantum Mechanics

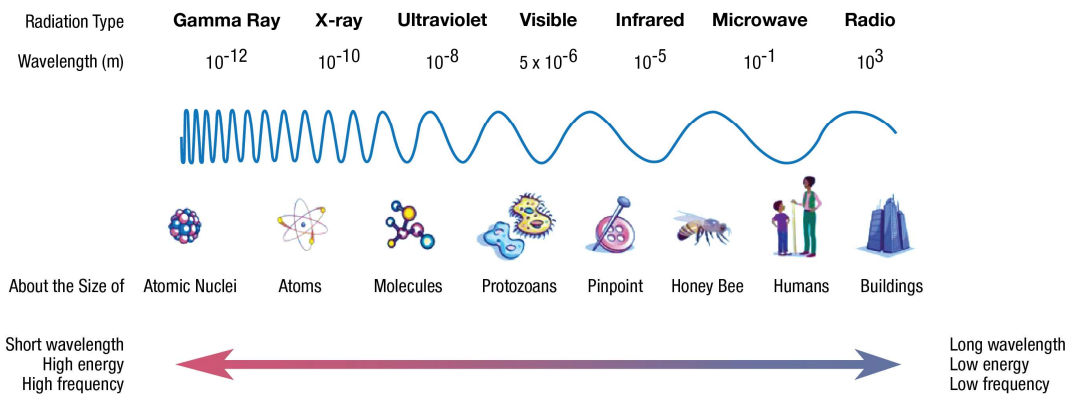
- Thermal Radiation
- Photoelectric Effect
- Compton Effect

Blackbody Radiation



Similarly, we receive light and heat from the Sun even without a medium (through a vacuum)

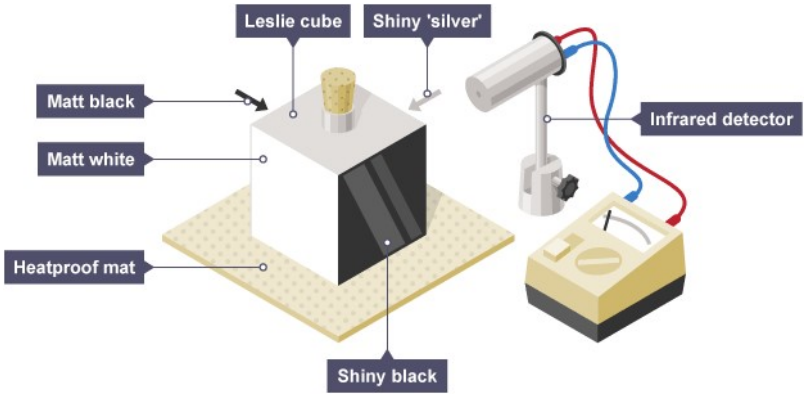
This type of radiation is called electromagnetic radiation



The rate of radiation energy emitted from a hot body depends on

1. The absolute temperature of the body (T in K)
2. The surface area (A)
3. The nature of the surface (ε)

This can be demonstrated using a Leslie cube



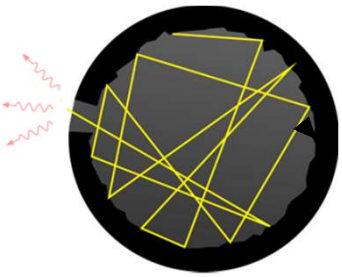
A Blackbody

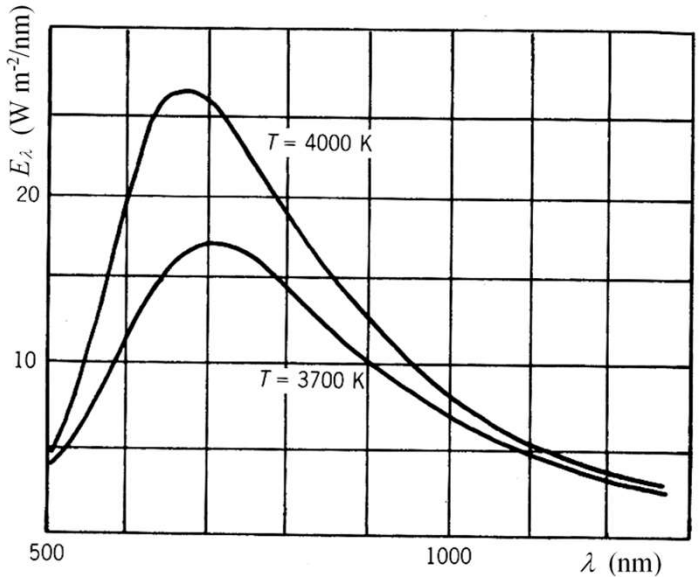
If all the electromagnetic radiation that falls on a body is absorbed, it is called a blackbody.

The blackbody is a perfect absorber as well as a perfect emitter.

By blackening the inner surface of a spherical shell with a hole made from a hard material, such as tungsten or ceramic, it can be used as a model blackbody.

Since all the e-m radiation entering from the hole is absorbed, it acts as a blackbody.



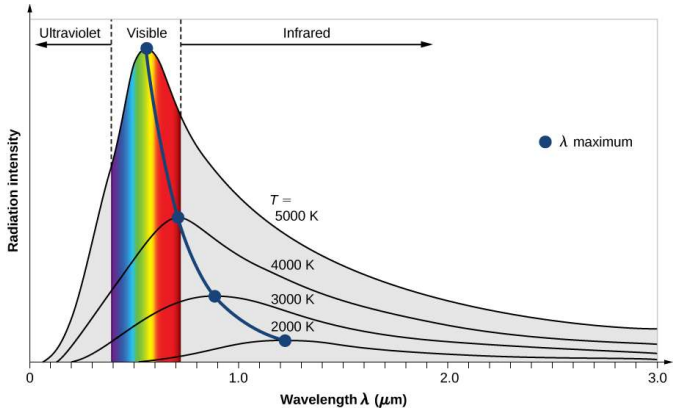


Wien's displacement law

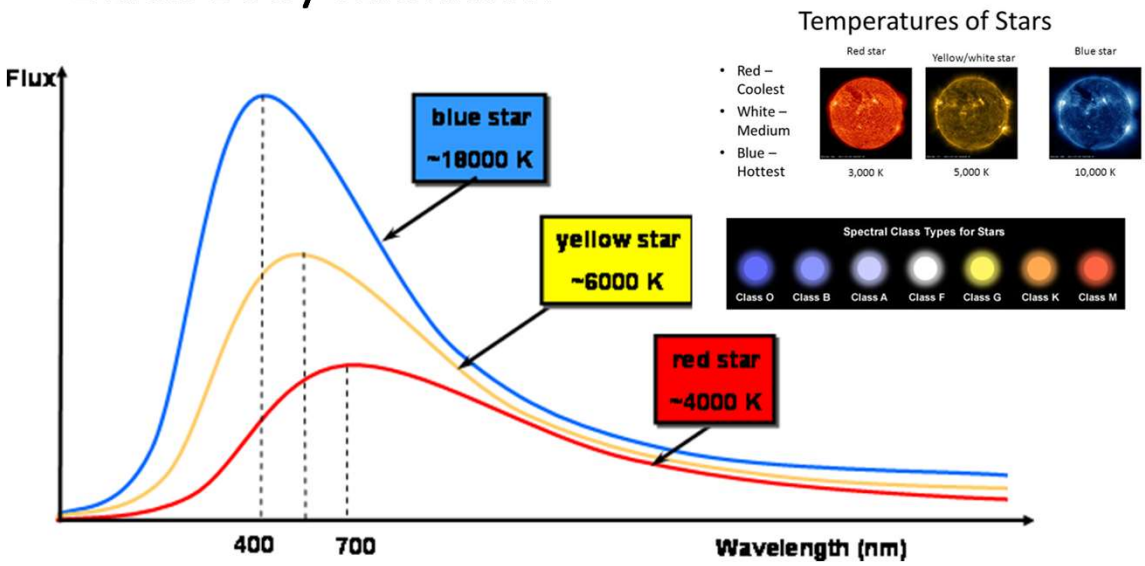
The product of the peak wavelength and the absolute temperature of the blackbody radiation curves is a constant.

$$\lambda_p . T = C$$

where C is the Wien's constant. $C = 2.90 \times 10^{-3}$ m K



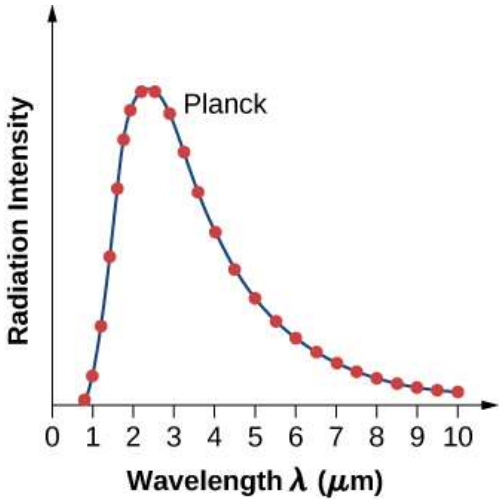
Black Body Radiation



Attempts to explain the blackbody radiation

The explanation of Rayleigh and Jean using classical physics theories did not match for shorter wavelengths

The explanation of Wein using classical physics theories did not exactly match for longer wavelengths



Quantum Hypothesis

The explanation of Max Panck using his **quantum theory** fits exactly the experimental data !

Planck's Hypothesis

Max Plank proposed Quantum hypothesis in 1900 concerning the nature of the oscillating molecules of the cavity walls of a blackbody.

1. The oscillating molecules that emit radiation could only have discrete energy levels given by

$$E = nh\nu$$

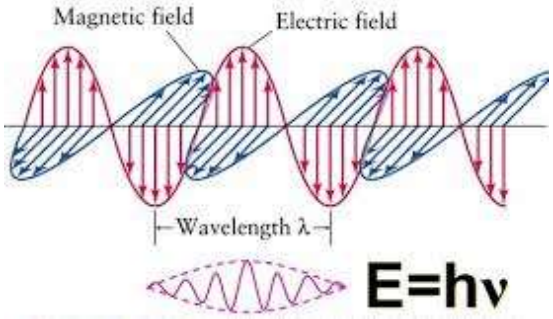
where n is a positive integer call a quantum number
ν is the frequency of oscillations
h is the Planck's constant ($h = 6.63 \times 10^{-34}$ J s)

2. The molecules emit or absorb energy in discrete value called quanta (or photons). Energy of a photon is given by

$$E = h\nu$$

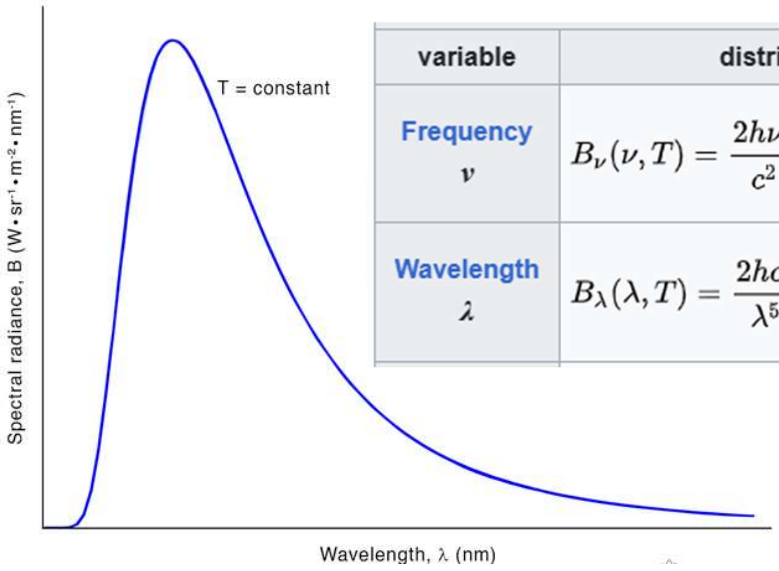


$E=h\nu$



E=hν

Planck's Law



variable	distribution
Frequency ν	$B_\nu(\nu,T)=\frac{2h\nu^3}{c^2}\frac{1}{e^{h\nu/(k_\mathrm{B}T)}-1}$
Wavelength λ	$B_\lambda(\lambda,T)=\frac{2hc^2}{\lambda^5}\frac{1}{e^{hc/(\lambda k_\mathrm{B}T)}-1}$